

WEEK-2

Program:

```
-- Step 1: Create Database
CREATE DATABASE Donation_db;
USE Donation_db;
-- Step 2: Create Books Table
CREATE TABLE Books(
    book_id INT PRIMARY KEY,
    title VARCHAR(100) NOT NULL,
    isbn VARCHAR(20) UNIQUE,
    published_year INT CHECK(published_year < 2027)
);
-- Step 3: Insert Records into Books Table
INSERT INTO Books VALUES
(1, 'The Great Gatsby', '9780743273565', 1925),
(2, 'To Kill a Mockingbird', '9780061120084', 1960),
(3, '1984', '9780451524935', 1949);
```

Output:

Output					
Action Output					
#	Time	Action	Message	Duration / Fetch	
✓ 1	13:30:09	CREATE DATABASE Donation_db	1 row(s) affected	0.016 sec	
✓ 2	13:30:11	USE Donation_db	0 row(s) affected	0.000 sec	
✓ 3	13:30:13	CREATE TABLE Books(book_id INT PRIMARY KEY, title VAR...	0 row(s) affected	0.047 sec	
✓ 4	13:30:15	INSERT INTO Books VALUES (1,'The Great Gatsby','978074327356...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec	

Program:

```
-- Step 4: Display Books Table
SELECT * FROM Books;

-- Step 5: Create Members Table
CREATE TABLE Members(
    member_id INT PRIMARY KEY,
    full_name VARCHAR(100),
    email VARCHAR(100) UNIQUE
);

-- Step 6: Insert Records into Members Table
INSERT INTO Members VALUES
(101,'Tejaswini','iliharshika@gmail.com'),
(102,'Bhanu Priya','bhanupriya@gmail.com'),
(103,'Sahasra','sahasra@gmail.com');
```

Output:

✓	5	13:30:49	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
✓	6	13:31:01	CREATE TABLE Members(member_id INT PRIMARY KEY, f...	0 row(s) affected	0.078 sec
✓	7	13:31:05	INSERT INTO Members VALUES (101,'Tejaswini','liharshika@gm...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec

Result Grid				
	book_id	title	isbn	published_year
▶	1	The Great Gatsby	9780743273565	1925
	2	To Kill a Mockingbird	9780061120084	1960
	3	1984	9780451524935	1949
•	NULL	NULL	NULL	NULL

Program:

```
SELECT * FROM Members;
```

Output:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Cont

	member_id	full_name	email
▶	101	Tejaswini	ilharshika@gmail.com
	102	Bhanu Priya	bhanupriya@gmail.com
	103	Sahasra	sahasra@gmail.com
*	NULL	NULL	NULL

Members 4 x | Apply | Revert | Context Help | Snippets

Output

Action Output

#	Time	Action	Message	Duration / Fetch
✓ 5	13:30:49	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
✓ 6	13:31:01	CREATE TABLE Members(member_id INT PRIMARY KEY, f...	0 row(s) affected	0.078 sec
✓ 7	13:31:05	INSERT INTO Members VALUES (101,'Tejaswini','ilharshika@gm...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
✓ 8	13:31:08	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
✓ 9	13:32:25	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
✓ 10	13:32:52	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Program:

```

31  -- Step 8: Create Loans Table
32  ● CREATE TABLE Loans(
33      loan_id INT PRIMARY KEY,
34      member_id INT,
35      book_id INT,
36      loan_date DATE,
37      FOREIGN KEY(member_id) REFERENCES Members(member_id),
38      FOREIGN KEY(book_id) REFERENCES Books(book_id)
39  );
40  -- Step 9: Insert Records into Loans Table
41  ● INSERT INTO Loans VALUES
42      (1,101,1,'2025-01-05'),
43      (2,102,2,'2025-01-08'),
44      (3,103,3,'2025-01-10'),
45      (4,101,2,'2025-02-01'),
46      (5,102,1,'2025-02-05'),
47      (6,103,2,'2025-02-12'),
48      (7,101,3,'2025-03-01'),
49      (8,102,3,'2025-03-07'),
50      (9,103,1,'2025-03-15'),
51      (10,101,1,'2025-04-01');
52  -- Step 10: Display Loans Table
53  ● SELECT * FROM Loans;

```

Output:

The screenshot displays a database management tool interface. At the top, there is a toolbar with options like 'Result Grid', 'Filter Rows', 'Edit', 'Export/Import', and 'Wrap Cell Content'. Below the toolbar is a 'Result Grid' showing a table with columns: loan_id, member_id, book_id, and loan_date. The table contains 10 rows of data, with the last row showing NULL values. To the right of the grid is a sidebar with icons for 'Result Grid', 'Form Editor', 'Field Types', and 'Query Stats'. Below the grid, there is a tab labeled 'Loans 6' and buttons for 'Apply' and 'Revert'. The bottom section is titled 'Output' and shows a log of actions performed, including SELECT, CREATE TABLE, and INSERT statements, along with their execution times and messages.

loan_id	member_id	book_id	loan_date
1	101	1	2025-01-05
2	102	2	2025-01-08
3	103	3	2025-01-10
4	101	2	2025-02-01
5	102	1	2025-02-05
6	103	2	2025-02-12
7	101	3	2025-03-01
8	102	3	2025-03-07
9	103	1	2025-03-15
10	101	1	2025-04-01
NULL	NULL	NULL	NULL

#	Time	Action	Message	Duration / Fetch
9	13:32:25	SELECT * FROM Books LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
10	13:32:52	SELECT * FROM Members LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
11	13:33:39	CREATE TABLE Loans(loan_id INT PRIMARY KEY, member...	0 row(s) affected	0.062 sec
12	13:33:45	INSERT INTO Loans VALUES (1,101,1,'2025-01-05'), (2,102,2,'20...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.032 sec
13	13:33:55	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
14	13:35:04	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

Program:

```
SELECT
    m.full_name AS Member_Name,
    b.title AS Book_Title
FROM Loans l
INNER JOIN Members m
ON l.member_id = m.member_id
INNER JOIN Books b
ON l.book_id = b.book_id;
```

Output:

Result Grid			Filter Rows:	Export:	Wrap Cell Content:
	Member_Name	Book_Title			
▶	Tejaswini	The Great Gatsby			
	Tejaswini	To Kill a Mockingbird			
	Tejaswini	1984			
	Tejaswini	The Great Gatsby			
	Bhanu Priya	To Kill a Mockingbird			
	Bhanu Priya	The Great Gatsby			
	Bhanu Priya	1984			
	Sahasra	1984			
	Sahasra	To Kill a Mockingbird			
	Sahasra	The Great Gatsby			

Program:

```
SELECT
    published_year,
    COUNT(book_id) AS Total_Books
FROM Books
GROUP BY published_year;
```

Output:

The screenshot displays a database management interface. At the top, a 'Result Grid' tab is active, showing a table with two columns: 'published_year' and 'Total_Books'. The table contains three rows of data. To the right of the grid is a vertical toolbar with icons for 'Result Grid', 'Form Editor', 'Field Types', and 'Query Stats'. Below the grid, there are tabs for 'Context Help' and 'Snippets'. At the bottom, an 'Output' panel is visible, showing a log of database actions. The log includes a table with columns for '#', 'Time', 'Action', 'Message', and 'Duration / Fetch'. The log shows several successful queries, including the one being executed, which returned 3 rows.

	published_year	Total_Books
▶	1925	1
	1960	1
	1949	1

#	Time	Action	Message	Duration / Fetch
✓ 13	13:33:55	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
✓ 14	13:35:04	SELECT * FROM Loans LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec
✓ 15	13:35:35	SELECT m.full_name AS Member_Name, b.title AS Book_Titl...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 16	13:36:50	SELECT published_year, COUNT(book_id) AS Total_Books F...	3 row(s) returned	0.000 sec / 0.000 sec
✓ 17	13:42:31	SELECT m.full_name AS Member_Name, b.title AS Book_Titl...	10 row(s) returned	0.000 sec / 0.000 sec
✓ 18	13:43:02	SELECT published_year, COUNT(book_id) AS Total_Books F...	3 row(s) returned	0.000 sec / 0.000 sec

Program:

```
-- Step 13: Create Donation History Table
CREATE TABLE Donation_History(
    donation_id INT PRIMARY KEY,
    book_id INT,
    donor_name VARCHAR(100),
    donation_date DATE,
    FOREIGN KEY(book_id) REFERENCES Books(book_id)
);

-- Step 14: Start Transaction
START TRANSACTION;

-- Step 15: Insert New Book into Books Table
INSERT INTO Books VALUES
(4,'Animal Farm','9780451526342',1945);

-- Step 16: Insert Donation Record and Commit Transaction
INSERT INTO Donation_History VALUES
(1,4,'Raj Kumar',CURDATE());

COMMIT;
```

Output:

✓	19	13:44:08	CREATE TABLE Donation_History(donation_id INT PRIMARY ...	0 row(s) affected	0.109 sec
✓	20	13:44:13	START TRANSACTION	0 row(s) affected	0.000 sec
✓	21	13:44:16	INSERT INTO Books VALUES (4,'Animal Fam','9780451526342'....	1 row(s) affected	0.032 sec
✓	22	13:44:21	INSERT INTO Donation_History VALUES (1,4,'Raj Kumar',CURDA...	1 row(s) affected	0.031 sec
✓	23	13:44:24	COMMIT	0 row(s) affected	0.016 sec

Program:

```
-- Step 17: Create Index on ISBN Column
CREATE INDEX idx_isbn
ON Books(isbn);
-- Step 18: Search Book Using ISBN
SELECT *
FROM Books
WHERE isbn='9780451524935';
```

Output:

The screenshot displays a database management interface. At the top, a 'Result Grid' shows the output of the SQL query. Below it, a vertical toolbar contains icons for 'Result Grid', 'Form Editor', 'Field Types', and 'Query Stats'. The bottom section, titled 'Output', shows a log of database actions with columns for '#', 'Time', 'Action', 'Message', and 'Duration / Fetch'.

book_id	title	isbn	published_year
3	1984	9780451524935	1949
*	NULL	NULL	NULL

#	Time	Action	Message	Duration / Fetch
20	13:44:13	START TRANSACTION	0 row(s) affected	0.000 sec
21	13:44:16	INSERT INTO Books VALUES (4,'Animal Farm','9780451526342',....	1 row(s) affected	0.032 sec
22	13:44:21	INSERT INTO Donation_History VALUES (1,4,'Raj Kumar',CURDA...	1 row(s) affected	0.031 sec
23	13:44:24	COMMIT	0 row(s) affected	0.016 sec
24	13:46:02	CREATE INDEX idx_isbn ON Books(isbn)	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.063 sec
25	13:46:08	SELECT * FROM Books WHERE isbn='9780451524935' LIMIT 0, ...	1 row(s) returned	0.000 sec / 0.000 sec